



RaySearch Laboratories AB (publ)
Olympiada Lachana
QA/RA Specialist
Eugeniavagen 18C
Stockholm, 113 68
Sweden

April 4, 2025

Re: K240398

Trade/Device Name: RayStation 2023B, RayPlan 2023B, RayStation 2024A, RayPlan 2024A,
RayStation 2024A SP3, RayPlan 2024A SP3

Regulation Number: 21 CFR 892.5050

Regulation Name: Medical Charged-Particle Radiation Therapy System

Regulatory Class: Class II

Product Code: MUJ

Dated: March 7, 2025

Received: March 7, 2025

Dear Olympiada Lachana:

We have reviewed your section 510(k) premarket notification of intent to market the device referenced above and have determined the device is substantially equivalent (for the indications for use stated in the enclosure) to legally marketed predicate devices marketed in interstate commerce prior to May 28, 1976, the enactment date of the Medical Device Amendments, or to devices that have been reclassified in accordance with the provisions of the Federal Food, Drug, and Cosmetic Act (the Act) that do not require approval of a premarket approval application (PMA). You may, therefore, market the device, subject to the general controls provisions of the Act. Although this letter refers to your product as a device, please be aware that some cleared products may instead be combination products. The 510(k) Premarket Notification Database available at <https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpmn/pmn.cfm> identifies combination product submissions. The general controls provisions of the Act include requirements for annual registration, listing of devices, good manufacturing practice, labeling, and prohibitions against misbranding and adulteration. Please note: CDRH does not evaluate information related to contract liability warranties. We remind you, however, that device labeling must be truthful and not misleading.

If your device is classified (see above) into either class II (Special Controls) or class III (PMA), it may be subject to additional controls. Existing major regulations affecting your device can be found in the Code of Federal Regulations, Title 21, Parts 800 to 898. In addition, FDA may publish further announcements concerning your device in the Federal Register.

Additional information about changes that may require a new premarket notification are provided in the FDA guidance documents entitled "Deciding When to Submit a 510(k) for a Change to an Existing Device" (<https://www.fda.gov/media/99812/download>) and "Deciding When to Submit a 510(k) for a Software Change to an Existing Device" (<https://www.fda.gov/media/99785/download>).

Your device is also subject to, among other requirements, the Quality System (QS) regulation (21 CFR Part 820), which includes, but is not limited to, 21 CFR 820.30, Design controls; 21 CFR 820.90, Nonconforming product; and 21 CFR 820.100, Corrective and preventive action. Please note that regardless of whether a change requires premarket review, the QS regulation requires device manufacturers to review and approve changes to device design and production (21 CFR 820.30 and 21 CFR 820.70) and document changes and approvals in the device master record (21 CFR 820.181).

Please be advised that FDA's issuance of a substantial equivalence determination does not mean that FDA has made a determination that your device complies with other requirements of the Act or any Federal statutes and regulations administered by other Federal agencies. You must comply with all the Act's requirements, including, but not limited to: registration and listing (21 CFR Part 807); labeling (21 CFR Part 801); medical device reporting (reporting of medical device-related adverse events) (21 CFR Part 803) for devices or postmarketing safety reporting (21 CFR Part 4, Subpart B) for combination products (see <https://www.fda.gov/combination-products/guidance-regulatory-information/postmarketing-safety-reporting-combination-products>); good manufacturing practice requirements as set forth in the quality systems (QS) regulation (21 CFR Part 820) for devices or current good manufacturing practices (21 CFR Part 4, Subpart A) for combination products; and, if applicable, the electronic product radiation control provisions (Sections 531-542 of the Act); 21 CFR Parts 1000-1050.

All medical devices, including Class I and unclassified devices and combination product device constituent parts are required to be in compliance with the final Unique Device Identification System rule ("UDI Rule"). The UDI Rule requires, among other things, that a device bear a unique device identifier (UDI) on its label and package (21 CFR 801.20(a)) unless an exception or alternative applies (21 CFR 801.20(b)) and that the dates on the device label be formatted in accordance with 21 CFR 801.18. The UDI Rule (21 CFR 830.300(a) and 830.320(b)) also requires that certain information be submitted to the Global Unique Device Identification Database (GUDID) (21 CFR Part 830 Subpart E). For additional information on these requirements, please see the UDI System webpage at <https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance/unique-device-identification-system-udi-system>.

Also, please note the regulation entitled, "Misbranding by reference to premarket notification" (21 CFR 807.97). For questions regarding the reporting of adverse events under the MDR regulation (21 CFR Part 803), please go to <https://www.fda.gov/medical-devices/medical-device-safety/medical-device-reporting-mdr-how-report-medical-device-problems>.

For comprehensive regulatory information about medical devices and radiation-emitting products, including information about labeling regulations, please see Device Advice (<https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance>) and CDRH Learn (<https://www.fda.gov/training-and-continuing-education/cdrh-learn>). Additionally, you may contact the Division of Industry and Consumer Education (DICE) to ask a question about a specific regulatory topic. See

the DICE website (<https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance/contact-us-division-industry-and-consumer-education-dice>) for more information or contact DICE by email (DICE@fda.hhs.gov) or phone (1-800-638-2041 or 301-796-7100).

Sincerely,

A handwritten signature in black ink, appearing to read "Lora Weidner". A large, light blue "FDA" watermark is visible in the background behind the signature.

Lora D. Weidner, Ph.D.
Assistant Director
Radiation Therapy Team
DHT8C: Division of Radiological
Imaging and Radiation Therapy Devices
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Office of Product Evaluation and Quality
Center for Devices and Radiological Health

Enclosure

Indications for Use

510(k) Number (if known)

K240398

Device Name

RayStation/RayPlan 2024A SP3, 2024A, 2023B

Indications for Use (Describe)

RayStation is a software system for radiation therapy and medical oncology. Based on user input, RayStation proposes treatment plans. After a proposed treatment plan is reviewed and approved by authorized intended users, RayStation may also be used to administer treatments.

The system functionality can be configured based on user needs.

Type of Use (Select one or both, as applicable)

☒ Prescription Use (Part 21 CFR 801 Subpart D)

☐ Over-The-Counter Use (21 CFR 801 Subpart C)

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K240398



510(K) SUMMARY (21 CFR § 807.92)

I. Contact Details

Applicant

RaySearch Laboratories AB (publ)

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Contact

Ms. Olympiada Lachana
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Preparation Date

4 April, 2025

II. Device

Device Trade Name

Device Name	Software Version number
RayStation	2024A SP3
RayPlan	2024A SP3
RayStation	2024A
RayPlan	2024A
RayStation	2023B
RayPlan	2023B

Common Name

Medical charged-particle radiation therapy system

Classification Name

System, Planning, Radiation Therapy Treatment

Regulation Number

892.5050

Product Code(s)

MUJ

III. Legally Marketed Predicate Device

Predicate #

K222312

Predicate Trade Name

RayStation 12A

Product Code

MUJ

IV. Device Description Summary

Explanation of how the device functions

RayStation is a software system for radiation therapy and medical oncology. Based on user input, RayStation proposes treatment plans. After a proposed treatment plan is reviewed and approved by authorized intended users, RayStation may also be used to administer treatments.

The system functionality can be configured based on user needs.

RayStation consists of multiple applications:

- The main RayStation application is used for treatment planning.
- The RayPhysics application is used for commissioning of treatment machines to make them available for treatment planning and used for commissioning of imaging systems.

The devices to be marketed, RayStation/RayPlan 2024A SP3, 2024A and 2023B, contain modified features compared to last cleared version RayStation 12A including:

- Improved sliding window VMAT (Volumetric Modulated Arc Therapy) sequencing
- Higher dose grid resolution for proton PBS (Pencil Beam Scanning)
- Automated field in field planning
- LET optimization (Linear Energy Transfer)

Device design information

These applications are built on a software platform, containing the radiotherapy domain model and providing GUI, optimization, dose calculation and storage services. The platform uses three Microsoft SQL databases for persistent storage of the patient, machine and clinic settings data.

As a treatment planning system, RayStation aims to be an extensive software toolbox for generating and evaluating various types of radiotherapy treatment plans. RayStation supports a wide variety of radiotherapy treatment techniques and features an extensive range of tools for manual or semi-automatic treatment planning.

The RayStation application is divided in modules, which are activated through licensing. A simplified license configuration of RayStation is marketed as RayPlan.

Feature	Description	Present in RayPlan
3D visualization	Displays the patient geometry and structures in three dimensions, with the possibility to rotate the patient image. If available, the dose distribution and beam modifiers are shown as well.	Yes
Adaptive replanning	The process of replanning the treatment for a patient, based on information about e.g. patient geometry, biology and dose delivery acquired during treatment.	No
Beam commissioning	Modeling of the radiation beam using a limited set of measurements on the clinical beam for commissioning treatment machines to make them available for treatment planning.	Yes
Beam design	Definition of beam orientations, apertures and various beam modifiers in order to manually create a treatment plan.	Yes
Beam set-up	Manual or automatic definition of isocenter, selection of treatment unit from the set of commissioned treatment machines, and specification of gantry/couch/collimator angles.	Yes
Beam's eye view	Displays the beam's eye view of the patient structures, fluence and beam modifier settings for any beam.	Yes
Brachy planning	Tools for planning of HDR brachytherapy treatments. Includes channel reconstruction and optimization and editing of dwell times.	Yes
CyberKnife planning	CyberKnife planning is completely integrated in RayStation. This includes optimization of high quality treatment plans collimated with MLC, fixed cones or iris cones, as well as support for all CyberKnife Synchrony techniques for target tracking and real time motion synchronization.	Yes
Deformable registration	Establishing a point-to-point mapping between two images using a deformation model. Used for mapping of dose and structures between images.	No
DICOM RT export	Export of images, structure set, plan, and dose according to the DICOM RT standard.	Yes

DICOM RT import	Import of images, structure set, plan, and dose according to the DICOM RT standard.	Yes
Dose calculation electrons	For electron beams RayStation calculates dose by the Monte Carlo technique. The electron beam phase space is generated in run time by sampling from a phase space model where the electrons are created at the secondary scattering foil. Electron transport towards the patient and energy transport and scoring in the patient is done using the Monte Carlo algorithm.	Yes
Dose calculation photons	For photon beams RayStation calculates dose by the point kernel superposition method (a.k.a. Collapsed Cone) or a Monte Carlo algorithm for radiation transport (8.1 onwards). The incident energy fluence is modeled as a superposition of a primary energy fluence and a scatter energy fluence. The dose contribution from contamination electrons is calculated by a pencil beam algorithm.	Yes
Dose calculation proton	For proton beams RayStation uses either the pencil beam algorithm with the Fermi-Eyges formalism, or a Monte Carlo algorithm for radiation transport (6.0 onwards). For passive beams the beam model accounts for the collimator and compensator block. For scanning beams the beam model accounts for the spot phase space including effects of air-scatter and beam paths through magnetic deflection elements. The user defined block aperture is taken into account in spot selection and optimization. In addition to this the relative biological effect (RBE) of proton beams is taken into account, resulting in a photon equivalent dose (8.1 onwards).	No
Dose calculation brachy	For brachy plans RayStation calculates dose based on the TG43 formalism.	Yes
Dose display (2D)	Displays the patient geometry with structures superimposed on the image data together with the dose distribution in transversal, sagittal, and coronal directions.	Yes

Dose tracking	Dose tracking scenarios including deformable registration of one CT or CBCT to another and subsequent deformation and accumulation of dose.	No
Eye planning	Tools for specifying a highly detailed geometrical model of the eye based on measurements from ultrasound and surgery. Support for positioning of tantalum clips. Import and visualization of fundus images. Creation and dose computation of proton plans with gaze angle based treatment directions	No
Fallback planning	Automatic generation of fallback plans using alternative treatment machines and treatment techniques. User-defined protocols specifies the setup of the fallback plans which are automatically generated from the protocols and optimized using dose mimicking functions.	No
Image conversion	Conversion of CBCT images to synthetic CT images that can be used for more accurate dose calculations.	No
Inverse planning	The user can define optimization settings such as optimization tolerance and maximum number of iterations as well as segmentation settings on the multileaf collimator and the Pencil Beam Scanning spot pattern. An interface for controlling the optimization process is provided and the progress of optimization is displayed in a view. The system generates control points for step-and shoot MLC plans, Sliding Window plans (DMLC), rotational plans (VMAT), 3DCRT plans, Wave Arc plans, TomoTherapy plans (6.1 onwards), proton, Pencil Beam Scanning plans, using the defined optimization problem. The inverse planning can be carried out either through a conventional inverse approach or by using multi-criteria optimization (photons and protons only).	Yes, but not for Wave Arc plans or proton planning.
LET evaluation	Computation and evaluation of dose-averaged linear energy transfer for proton plans. LET is an additional physical quantity that can be used to assess the radiobiological effect of the proton radiation.	No
LET optimization	Possibility to include optimization functions on the dose-averaged linear energy transfer in addition to the dose for proton PBS.	No

Machine database	Microsoft SQL database for storage of beam model parameters, machine constraints and dose curves with dosimetric data for treatment units.	Yes
MR based planning	Allowing MR-images as planning images and base dose computation on material override ROIs.	Yes
Optimization functions	The optimization functions are specified in terms of objectives and constraints to form the optimization problem that is solved by the optimization engine.	Yes
Patient anatomy modeling	Manual and semi-automatic segmentation tools for contouring ROIs slice by slice together with semi-automated generation of the patient outline ROI. The model-based segmentation technique allows for semi-automatic delineation of structures by matching 3D shape models of the structures to new image data. With atlas-based segmentation, the user can define templates consisting of already segmented image data and use this template for segmentation of new patient images. With deep learning segmentation, the user can use trained deep learning models for automatic segmentation of new patient images. (The model training is performed offline on clinical CT and structure data.)	Yes, but not for atlas based segmentation or deep learning segmentation.
Patient database	Microsoft SQL database for storage of all patient and plan data. Not for long term storage.	Yes
Plan Explorer	The system computes a large set of plans according to given rules and the user is provided with tools to select good plans from these.	No
Proton arc planning	Generation of rotational proton arc plans. Support for discrete PBS arcs.	No
Quality assurance preparation	Tools for transferring the clinical plan to a phantom and recalculate dose. The output is the dose distribution in DICOM format or a 2D dose plane and a QA report. Predicted EPID response is retrieved by photon dose computation in a specially designed phantom.	Yes

RBE dose handling	RBE (Relative Biological Effectiveness) models can be defined and commissioned. For proton treatments (8.1 onwards), the user can select whether to look at RBE-corrected dose or physical dose. Dose summation is only possible for photon doses and RBE-corrected proton doses.	No
Robust evaluation	Tools used to answer questions of how the dose distribution would appear if the patient setup at the time of treatment does not fully correspond to the planning CT. A model of patient uncertainties such as CT inaccuracy and setup errors is used to compute a set of scenario doses for evaluation. Up to version 8.1 the support was limited to computation of one scenario at a time in Plan Evaluation.	No
Robust optimization	Optimization where a model of patient uncertainties such as CT inaccuracy, setup errors or organ motion is used during the optimization.	No
Scripting	Scripting gives programmatic access to functionality, excluding user risk mitigations. Through scripting, the clinic specific procedures can be automated. The operating system and other applications can be accessed.	No
Supported treatment positions	HFS, FFS, HFP, FFP	Yes
	Decubitus left/right	Yes
	Seated position (for proton)	No
System integrity tools	Hardware based license, preventing unauthorized useable copies to be made. Checksum control of binary files to prevent tampering. Data in the patient and machine databases only available for users with administrator rights.	Yes
TomoTherapy planning	Planning for TomoTherapy machines is completely integrated in RayStation. Also provides tools for selection of targets and imaging angles for the TomoTherapy machine to use for target tracking during delivery.	Yes

	Supported techniques: TH = TomoHelical, TD = TomoDirect	
Treatment adaptation	A general concept where the treatment plan is adapted during the course of treatment. Tools available today include deformable dose accumulation, CBCT dose calculation and replanning scenarios.	No
Treatment plan approval	Approval of the preferred treatment plan and referenced ROIs by authorized medical staff. Once a treatment plan is approved, it is locked for any further modification.	Yes
Treatment plan creation	Treatment plan creation with specification of plan properties such as number of fractions and delivery technique.	Yes
Treatment plan evaluation	Evaluation of a single plan. Comparison of dose distributions and DVH curves of two or three plans.	Yes
Undo/redo and auto recovery	The undo stack is saved to the database, enabling recovery of RayStation after crash. The user may redo all or selected changes at reopen of patient after crash.	Yes
Virtual Simulation	Setup of isocenter, beam arrangements and basic aperture design. Export to laser systems for patient marking.	Yes

Scientific concepts that form the basis for the device and significant performance characteristics:

RayStation is a stand-alone software medical device intended for radiation therapy. Input to the device is patient, disease and treatment unit information, output from the device is one or more treatment plans. The treatment plans include treatment unit parameter settings for optimal beam arrangements, energies, field sizes, and ultimately fluence patterns to produce a safe and effective radiation dose distribution, as compared to the predicate.

The scientific concepts of a treatment planning system are patient and beam modeling, and algorithms for dose calculation and plan parameter optimization.

The patient model is a computerized representation of the patient tissue and densities, identifying the target regions and particular organs at risk. The model is based on medical images of the patient and must have the desired level of accuracy. Likewise, the beam modeling is a computerized representation of the treatment unit, defined by fluence type, energy distribution, machine specific geometry, and beam modifiers such as MLC, flattening filters, wedges etc. The algorithms for dose calculation and plan parameter optimization must take into account all geometries and materials that affect irradiation transport through the treatment unit and the patient. The optimization

algorithm iterates treatment plan parameters until the desired treatment plan and dose distribution have been obtained. Also here, all steps must be done to the desired level of accuracy.

Significant physical characteristics of the device, material used, and physical properties:

The device is a standalone software medical device. It has no physical properties or materials. The device design information can be found in the subsection above “Device design information”.

V. Intended Use/Indications for Use

RayStation is a software system for radiation therapy and medical oncology. Based on user input, RayStation proposes treatment plans. After a proposed treatment plan is reviewed and approved by authorized intended users, RayStation may also be used to administer treatments.

The system functionality can be configured based on user needs.

Indications for Use Comparison

The indications for use are the same as the predicate device.

VI. Technological Comparison

Comparing RayStation 2024A SP3, 2024A and 2023B with RayStation 12A, all devices are treatment planning systems. Based on user input, all three RayStation versions propose treatment plans. After a proposed treatment plan is reviewed and approved by authorized intended users, all three RayStation versions may also be used to administer treatments.

General features comparison table

Item	Compared to RayStation 12A	Comment
Hardware platform	Substantially Equivalent	RayStation 2024A SP3, 2024A, 2023B and 12A all use standard office PCs as hardware platform.
Operating system	Substantially Equivalent	RayStation 2024A SP3, 2024A, 2023B and 12A all use Windows 10 Professional (or higher) and Windows Server 2012 R2 (or higher). See RayStation 2024A System Environment Guidelines for details.
Target population	Substantially Equivalent	RayStation 2024A SP3, 2024A, 2023B and 12A are intended for the same target population and anatomical sites; persons that have been prescribed an external beam radiation therapy or chemotherapy treatment.
Anatomical sites	Substantially Equivalent	
Human factors	Substantially Equivalent	In terms of human factors, the systems are considered equivalent. Minor updates only. The user interfaces use the same framework with the same controls and views.
Standards met	Substantially Equivalent	RayStation 2024A SP3, 2024A, 2023B and 12A all comply with the IEC 61217, IEC 62083, IEC 62304, IEC 62366-1 and ISO 14971.

Item	Compared to RayStation 12A	Comment
Image types	Substantially Equivalent	RayStation 2024A SP3, 2024A, 2023B and 12A all support CT, PET and MR images for identifying patient organs and contouring.
Reporting aspects	Substantially Equivalent	When evaluating and approving treatment plans, all necessary data is presented to the user and available in print in all systems.
Image storing	Substantially Equivalent	None of the systems are intended for long term storage of images or other patient data.
Network / remote connections and capabilities	Substantially Equivalent	All systems are capable of network transfer of patient data using the DICOM protocol. RayStation 2024A SP3, 2024A, 2023B and 12A are designed for desktop use and for remote access using standard virtualization techniques. Remote connection to the system is verified in detail and equivalent to local connection.
Cybersecurity	Substantially Equivalent	No architectural changes or major new features that affect cyber security.

The above listed differences in the general features do not raise different questions of safety or effectiveness.

Some software features and characteristics in RayStation 2024A SP3, 2024A and 2023B are different from the predicate device, RayStation 12A. However, these differences are considered by RaySearch to be enhancements of the predicate. The principle of operation of the subject devices is the same as that of the existing predicate device. Verification and validation demonstrate that the subject devices are as safe and effective as the predicate. RaySearch therefore believes that the subject devices are substantially equivalent to the predicate device.

The comparison of the added/updated functions in RayStation 2024A SP3, 2024A and 2023B compared with predicate device, RayStation 12A is presented in the table below.

Added/updated function in RayStation 2024A SP3, 2024A and 2023B compared with predicate device, RayStation 12A.	Description of the RayStation function	Compared to RayStation 12A K222312
Dose compensation point computation for Tomo Synchrony	A new way of computing the dose compensation point was added since the previously exported position was suboptimal. The point is calculated by taking the center of the volume formed of the dose voxels with dose above 90% of the max dose.	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness
RayStation does not populate DoseCompensationPoint correctly for TomoMotion plans	Related to the item above. The DICOM export was updated so that the dose compensation point coordinates in IDMS are populated with the correct coordinates (as defined in the related item above).	Substantially Equivalent. The feature does not raise different questions of

Added/updated function in RayStation 2024A SP3, 2024A and 2023B compared with predicate device, RayStation 12A.	Description of the RayStation function	Compared to RayStation 12A K222312
		safety or effectiveness
Most distal energy layer missing causing ripples in dose distribution for box target in water phantom	Ripples sometimes occurred in the distal edge of Pencil Beam Scanning dose distributions for box targets in water phantoms in combination with some dose grid setups.	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness
Improved replanning	The improvement to adaptive replanning is to remove dialog options to simplify the workflow and to allow the user to select whether to include background dose or not. Adaptive replanning allows the user to adapt the treatment to a changing patient geometry. Reasons for when a patient is selected for replanning can vary between treatment protocols. Replanning should be performed when the plan is no longer suitable for treatment and when it has become necessary to adapt the plan to the new circumstances. The prerequisite for adaptive replanning in RayStation is an approved treatment plan. The approved treatment plan that is selected for adaptation is called the base plan. The new adapted plan will be a copy of the base plan. In previous versions of RayStation there was a separate dialog for creation and editing of adapted plans. The New/Edit adapted plan dialog displayed all plan parameters copied from the base plan, and the parameters (such as treatment machine) were editable. To simplify the plan adaptation workflow with less options, and to simplify the code design, the Create adaptive plan dialog in the new version of RayStation only provides some basic plan parameters such as plan name and selection of planning image set. If the user wants to change other parameters, the plan can be edited using the regular Edit plan dialog after creation. Also, with previous versions of RayStation, adapted plans were always made taking background dose into account. Either the accumulated delivered dose for previous fractions (computed in the Dose tracking module) or the planned dose for a number of fractions. The background dose was considered when optimizing the adapted plan on the new image set.	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness

Added/updated function in RayStation 2024A SP3, 2024A and 2023B compared with predicate device, RayStation 12A.	Description of the RayStation function	Compared to RayStation 12A K222312
	The new RayStation version makes it possible also to create adapted plans without taking any background dose into account. This change allows a quick and simple replanning workflow where a base plan is quickly adapted to the daily patient geometry.	
Improved dose tracking	The Dose tracking module in RayStation is used to evaluate the actual delivered dose to the patient to evaluate daily changes as well as to evaluate trends as the treatment progresses. With previous versions of RayStation it has been possible to do dose tracking for one selected treatment plan. For each fraction it was possible to import and to compute dose for the plan selected for dose tracking on the treatment time image of the patient. When considered needed a new plan, based on the original plan but adapted to a new image of the patient, could be made and be used as basis for subsequent dose tracking. With the new version it is possible to select which treatment plan or beam set to use for dose tracking per fraction, to be able to fully simulate the actual treatment. In several scenarios you treat with different plans on different fractions, e.g. based on bladder filling. With the new version it is also possible to add and remove fractions to the treatment course. This limited change in RayStation makes the dose tracking functionality more useful since they can be used in more treatment scenarios.	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness
Delete ROI contours in all slices except every n:th	This change increases usability when the user manipulates patient image contours. When editing existing region of interests (ROIs) it sometimes is efficient to delete contours on certain image slices and to let RayStation's interpolation algorithm create new contours on those slices. In previous versions of RayStation the user could delete a complete ROI geometry or one contour at a time. With the new version of RayStation, contours in several slices for the selected ROI can be deleted, keeping contours in e.g. every 2nd, 3rd or 5th slice. Optionally, it is possible to define a limited range of image slices within which to do this.	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness
Allow voxelwise min/max doses if scenario group contains a single image set Scripting access to voxelwise min/max dose	This change enables an evaluation criterion for more scenarios and brings scripting closer to what is already enabled for the user through the device graphical user interface.	Substantially Equivalent. The feature does not raise different questions of

Added/updated function in RayStation 2024A SP3, 2024A and 2023B compared with predicate device, RayStation 12A.	Description of the RayStation function	Compared to RayStation 12A K222312
distributions in robust evaluation	Possibility to compute voxelwise min/max doses as long as all scenarios are on a single image set. The image set is no longer restricted to being the nominal image. Scripting access to the clinical goal results with respect to the voxelwise min/max distributions for a scenario group. Scripting access to the actual voxelwise min/max dose distribution. These could not be accessed using scripting in previous versions.	safety or effectiveness
Various DICOM improvements	Two new treatment machine settings for export have been added in machine modeling to improve integration with Varian's Aria system. 1) Possibility to configure if jaws shall be exported as ASYMX/ASYMY or X/Y. X/Y is set in the exported file (instead of ASYMX/ASYMY) for machines that have this setting and all segments in all beams are symmetrical. 2) Setting that will remove MLC settings from export for cone plans, if all MLC positions are "withdrawn". The machine settings have been added for DICOM plans exported from RayStation to comply with Varian's Aria system. Added the possibility to define the default DICOM export target. This is useful to make the folder that is most often used for export, the folder selected by default. A new machine configuration for omitting pixel intensity relationship attributes in the export has been added for enhanced compatibility with other systems. In RayStation 11B, the attributes (0028, 1040) Pixel Intensity Relationship and (0028, 1041) Pixel Intensity Relationship Sign were added to the export of RT Images. The value 'OTHER' for Pixel Intensity Relationship Sign has recently been added to the DICOM standard and hence some older systems cannot handle that value. This new option makes it possible to exclude the attributes at export, when Pixel Intensity Relationship Sign = 'OTHER'. Added possibility to select if option "Delete original DICOM files after import" shall be default. This is useful to have the original DICOM files deleted by default after import.	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness
Point-dose optimization in brachy plans	Change to increase usability by allowing the user to work with optimization objectives and constraints for points of interest (POI)s in a way similar to what was already available for regions of interest (ROI)s.	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness

Added/updated function in RayStation 2024A SP3, 2024A and 2023B compared with predicate device, RayStation 12A.	Description of the RayStation function	Compared to RayStation 12A K222312
	It is now possible to add objectives and constraints on POIs. The supported function types are Min dose, Max dose and Target dose. Otherwise, the functions work similar to ROI functions. In this way it is possible to perform planning without delineating targets and OARs.	
Electron Monte Carlo dose engine improvements	Minor improvements to the new electron Monte Carlo dose engine version (introduced in RayStation 12A). The user noticeable updates are multi-GPU support and support for different shapes of collimator tips. Multi-GPU: For performance reasons, the user can select to distribute the computation on multiple GPUs if the computer running RayStation has more than one GPU card. Distributing the computation on multiple GPUs will make it faster. The computed dose is identical between one or multiple GPUs and thus accuracy is not affected. Different tip shapes: The user can specify in the machine model if the shape of the MLC and jaw tips should be focused or rounded. Previously, focused tips were always used. The selected tip shape is used during dose computation, giving a more accurate description of the physical machine.	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness
Linac machine parameter improvements	The changes are done to achieve more exact simulation in RayStation of how MLC leaves and jaws are positioned in the machine during delivery.	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness
Automated field in field planning	The new Auto field-in-field tool is used to create a field-in-field plan based on the primary prescription and a primary field that is either defined by a target plus a user defined margin or by the first segment in each beam. The algorithm starts with a primary field and subfields are added iteratively to irradiate low dose regions. The tool automatically: - creates subfields based on low dose regions - adjusts segment weights - computes final dose and scales to prescription The tool automates a workflow that previously has been possible to do manually.	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness
Segment weight optimization using photon Monte Carlo added	Optimizing with respect to only segment weights, called Segment MU in the RayStation user interface, were previously only available for one of the two clinical photon dose engines in RayStation, the Collapsed Cone dose engine. Now segment weight	Substantially Equivalent. The feature does not raise different questions of

Added/updated function in RayStation 2024A SP3, 2024A and 2023B compared with predicate device, RayStation 12A.	Description of the RayStation function	Compared to RayStation 12A K222312
	optimization is possible also when using photon Monte Carlo, the other clinical photon dose engine in RayStation. In the RayStation user interface, it's possible to select to optimize with respect to Segment shapes and/or Segment MU. In 12A however, it was not allowed to start Segment MU only optimization with photon Monte Carlo selected. With photon Monte Carlo selected in 12A, it was only possible to optimize Segment shapes individually or Segment shapes together with Segment MU. With Collapsed Cone selected, it was possible in 12A to optimize Segment shapes individually, Segment MU individually or Segment shapes together with Segment MU. In 2023B all options are available with photon Monte Carlo selected as well. This change makes it possible to utilize the more accurate Monte Carlo dose engine in more planning scenarios.	safety or effectiveness
Photon Monte Carlo dose engine: Improved positron handling	The photon Monte Carlo dose engine has been improved with respect to positron physics. For external beam treatment energies, the difference in computed dose is small, but remodeling of photon and electron Monte Carlo machine models is required. The most noticeable change in computed dose before remodeling is the output for larger field sizes for photon Monte Carlo. There are no changes to the RayStation user interface. This change further improves the accuracy of computed photon Monte Carlo dose.	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness
Improvements to conformity of broad beams when using compensator smearing/gradient	For proton broad beam techniques, manual editing of the compensator shape is sometimes done to optimize the dose distribution. Clinics reported that for some plans where smearing was applied after manual compensator editing, the modulation got insufficient to cover the proximal edge of the target. For proton broad beam techniques, "Compute beam SOBP" (spread out Bragg peak) now traces through the actual shape of the compensator and proton wedge (if present). This change makes it easier to get a good conformance to the target after manually editing the compensator or applying smearing and gradient tools.	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness
LET optimization	Linear energy transfer (LET) is a quantity that, in combination with the dose, provides information about the biological effect of the radiation. Typically, in risk organs you want to avoid high LET in combination with high dose. High LET with low dose is more	Substantially Equivalent. The feature does not raise different questions of

Added/updated function in RayStation 2024A SP3, 2024A and 2023B compared with predicate device, RayStation 12A.	Description of the RayStation function	Compared to RayStation 12A K222312
	acceptable. Areas of the body that receive high dose and high LET can experience more cell death, which is desirable in the tumor, but harmful to normal tissue. The new version of RayStation supports optimization on dose-averaged linear energy transfer (LETd) for protons. This can be used to lower LET in risk organs, which could reduce potential side effects of radiotherapy. Possibility to add Max LETd and Min LETd optimization functions in addition to standard dose optimization functions has been added.	safety or effectiveness
Evaluation on converted CBCT images for protons	This is a change to enable users to visualize proton dose on CBCT images for evaluation only. The images cannot be used for planning and the dose cannot be used clinically. It is possible to compute an evaluation dose on converted CBCT images for protons. It is not possible to use such an image for planning. The conversion algorithms themselves are the same as those already clinically available for photons.	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness
Support for discrete proton arcs	PBS Arc is a new treatment technique in RayStation. However, since the arc plans are exported as regular PBS plans, with many fields per plan instead of many segments per field, this is an administrative change. RayStation 2023B includes optimization of discrete PBS arc plans where the gantry is stationary during dose delivery, as opposed to dynamic arc plans where the gantry moves during delivery. Discrete PBS arc optimization involves: - Many gantry angles per beam, where multiple energy layers are delivered per gantry angle. - Easy setup including air gap computation for collision avoidance. - Iterative reduction of energy layers during optimization to reduce delivery time. PBS arc plans must be converted into regular PBS plans with many beams, to be available for export out of RayStation.	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness
Separate clinical goals per beam set and plan	This change increases usability for evaluation of clinical goals. It is now possible to specify which dose to evaluate a specific clinical goal on. Clinical goals are clearly grouped by plan and beam set in the GUI.	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness

Added/updated function in RayStation 2024A SP3, 2024A and 2023B compared with predicate device, RayStation 12A.	Description of the RayStation function	Compared to RayStation 12A K222312
	In previous versions all clinical goals are evaluated on the currently selected dose (plan dose/beam set dose/beam dose/etc).	
Support and fixation structures per beam set	This change increases usability by not forcing all beam sets to have the same couch and fixation device settings. This is to support the possibility to use different couches and different fixation devices for different beam sets. (E.g. changing treatment machine often means changing to a different kind of couch.) Only selected fixation and support ROIs will be included in dose computation.	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness
One-click creation of synthetic CT	This change increases usability for creation of synthetic CT. In previous versions, to create a sCT using the CBCT correction algorithm or the virtual CT algorithm, the user had to provide a pre-created deformable registration between refCT and CBCT as well as a pre created Field of View (FOV) ROI on the CBCT. Also – an external was required on the CBCT. The new functionality allows sCT to be created from two rigidly registered refCT and CBCT.	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness
Modality of sCT shall be CT	This change increases the usability when handling CBCT images. The user can now freely choose the imaging system of a converted CBCT image. Converted CBCT images fetch their HU-to-mass density table from their imaging system rather than from the imaging system of the reference CT image. As a result, the modality of such images will correspond to that of the chosen imaging system which in most cases will be CT instead of CBCT.	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness
Various GUI improvements throughout the system	- Adjusted treatment delivery information in Patient data management module - ROI/POI lists initially sorted alphabetically - Sorting on sub-columns for some tables - Reduced size of aperture shapes toolbar - Show beam parts, Volume rendering settings and DRR settings dialogs are now non-modal - Improved performance of “Copy to all” of Visualization settings in ROI/POI details - Image conversion histogram updated to follow the color scheme in the fusion view - Show segment number in BEV - Possibility to translate also plan reports and the Report Designer	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness

Added/updated function in RayStation 2024A SP3, 2024A and 2023B compared with predicate device, RayStation 12A.	Description of the RayStation function	Compared to RayStation 12A K222312
Various workflow improvements	These are changes for increased usability. - Creating structures from template now have the option to automatically update derived ROIs. - Option to replace existing clinical goals when applying template. - Option to replace existing optimization functions when applying template.	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness
Various general system improvements	These are changes for increased usability. - Treat and Protect settings are now scriptable for all modalities. Notably for protons and electrons. - Possibility to copy objectives and constraints - Function values are no longer automatically computed after final dose - Relative dose value has been added to Dose cloud visualization - Faster saving of patient data - Faster Bragg Peak rendering - For DICOM data where a filter has been applied the Transfer Syntax Implicit VR Little Endian will always be used	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness
Material view refactoring	This change increases usability of the material visualization view. Make the material visualization view less error prone and possible to use with beam set specific support/fixation ROIs. This will lead to limitations in the current functionality; no material view will be shown when there is no computed dose, and the material visualization view will only be shown for beam set and beam doses. One extension to the current functionality is made; material will be visualized for Bolus ROIs for beams with the Bolus ROI assigned.	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness
Updated DRR UID handling	This is an administrative change for handling unique identifiers (UIDs). In the 2024A RayStation/RayCare releases, a joint goal is to be able to deliver treatment plans using Varian TrueBeam machines. In that interface, information about the RT Images/ Digitally Reconstructed Radiographs (DRRs) that will be used for delivery needs to be provided. To avoid incorrect RT Images being used at patient positioning, RayStation needs to produce deterministic UIDs for exported RT Images. The logic to generate DRRs has previously been spread out and duplicated in various places in RayStation. Now the modelling of DRRs will be centralized in the RayStation domain model which will both solve the immediate	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness

Added/updated function in RayStation 2024A SP3, 2024A and 2023B compared with predicate device, RayStation 12A.	Description of the RayStation function	Compared to RayStation 12A K222312
	issues and decrease the risk of introducing future problems when the DRR generation is updated.	
Dose rate for setup beams	This is a change to fulfill an administrative interface requirement since a dose rate for a setup beam does not have any function. Possibility to set a dose rate for RayStation setup beams when using a Linac treatment machine. The dose rate for setup beams is defined per Linac treatment machine in RayPhysics.	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness
Display brachy channel numbers in 3D view	This is a change to increase usability. It shall be possible to display channel numbers in the 3D view.	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness
Improved shape conversion	This is a change to increase performance and accuracy of a calculation. The specifications are the same and there is no design change. New algorithm for converting triangle mesh ROIs to voxel ROIs. Improves accuracy of the conversions at first and last slice of ROI and improves performance for cases with a lot of mesh ROIs.	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness
General patient modeling improvements	These are changes to increase the usability of patient modeling workflows. - ROIs grouping by body site in DLS dialog - Possibility to create ellipsoid ROIs - Default names for MBS ROIs now follow the TG263 standard - Improved non-uniform expansion/contraction of ROIs - Delete multiple contours (keeping every n:th) now works in all view directions - Possibility to add colors for ROIs in RayMachine	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness
Image registration improvements	This change improves the visualization of the image registration workspace. The 'Deformation grid' in the Deformable registration module view now shows image set in the same direction as the reference image set for easier comparison. - The floating view in the Image registration module has been updated and improved. It now works like it did in RayStation 11A and earlier versions. Position, Direction	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness

Added/updated function in RayStation 2024A SP3, 2024A and 2023B compared with predicate device, RayStation 12A.	Description of the RayStation function	Compared to RayStation 12A K222312
	(transversal/sagittal/coronal), Patient direction letters, Imaging system name and Slice number are again displayed in the floating view.	
Improved Varian TrueBeam integration	This is an administrative change to fulfill interface requirements from other devices. The new RayStation version supports indicating export of plans with high MU values as High-Dose Technique Type plans. The version also includes improved Dose Reference Description handling. In RayPhysics, it is possible to define MU thresholds for different treatment techniques. The thresholds are defined per treatment machine. When exporting a treatment plan, the DICOM tag (300A, 00C7) in the RTPlan is set to SRS for beams where MU exceeds the threshold. The population of the DICOM attributes Prescription Description and Dose Reference Description has been updated. Previously, default values were used to populate these attributes. For the Dose Reference Description, it is now possible to select between four different default modes for populating the values. This setting can be configured per machine. It is also possible to set user defined overrides for both attributes. These options are added to reduce the need to manually edit the properties after importing a plan into other systems.	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness
Improved sliding window VMAT sequencing	Control points are key elements in defining the delivery of a beam. They specify how the beam parameters change over the course of the treatment delivery. A beam typically contains multiple control points to describe the beam's modulation and movement. One important attribute of a control point is the gantry angle. It was found that for VMAT beams, the way RayStation used decimal distances between control points could lead to treatment plans that could not be delivered by the treatment machine (e.g. because they violated machine constraints such as maximum gantry speed) since the gantry angles had to be rounded to fewer decimals when the plan was exported to DICOM. In the new version the sliding window VMAT sequencing algorithm has been modified to create control points with a	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness

Added/updated function in RayStation 2024A SP3, 2024A and 2023B compared with predicate device, RayStation 12A.	Description of the RayStation function	Compared to RayStation 12A K222312
	gantry spacing of exactly 2 degrees, as opposed to a gantry spacing of maximum 2 degrees. This change ensures that the created VMAT beams are deliverable on the treatment machine.	
Varian Halcyon in EPID validation	This change enables the previously existing functionality to be used for one additional electronic portal imaging device (EPID). The EPID QA functionality has now also been validated for Varian Halcyon. EPID measurements using a Varian Halcyon LINAC have been added to the EPID QA validation. All validation measurements pass the acceptance criteria, and the license for EPID QA can be made available also for Varian Halcyon users.	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness
Possibility to use background proton dose computed on converted CBCT	With this change, a proton dose computed on a converted CBCT image set can also be used as background dose in optimization and in adapted plans. With the previous version, it was possible to compute evaluation dose on converted CBCT images for protons. The image conversion algorithms are the same as those already clinically available for photons.	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness
Higher dose grid resolution for proton PBS	In RayStation the dose grid defines the volume where dose is to be computed. The finer resolution the dose grid has, the more accurate is the computed dose. (The dose takes longer to calculate with a finer dose grid, and more data is created, so there are trade-offs between high resolution and performance.) Previous versions of RayStation had support for a dose grid resolution down to 1x1x1 mm. For treatments of small areas, it can be beneficial to be able to use very fine dose grids. Treatments of eye cancer is one such example. To better support these kinds of treatments RayStation now supports dose grid resolutions down to 0.5 mm when using the proton PBS Monte Carlo and Pencil Beam dose engines. This includes support for computing curves at this high resolution in RayPhysics, as well as for computing proton curves at non-uniform resolution in RayPhysics.	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness
Upgrade of CUDA	CUDA (Compute Unified Device Architecture) is a parallel computing platform and API developed by NVIDIA, enabling developers to leverage NVIDIA GPUs (Graphics Processing Units) for general-purpose processing. It enhances performance by allowing the execution of thousands of parallel threads. CUDA is used in several algorithms in RayStation that utilize the GPU, e.g., in dose engines and algorithms for deformable registration.	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness

Added/updated function in RayStation 2024A SP3, 2024A and 2023B compared with predicate device, RayStation 12A.	Description of the RayStation function	Compared to RayStation 12A K222312
	The platform used for GPU computations in RayStation (CUDA) has been upgraded to version 12.2 to support new hardware environments and compiler versions. The platform upgrade does not affect the use of GPU calculations with the treatment planning workflow.	
FSN/recall corrections	• RES 91867/FSN 109886: DICOM export from the Virtual Simulation module • RES 94153/FSN 130646 Remove template material 'Silicon [Si]' • RES 94388/FSN 133261: Incorrect early exit in SSD • RES 96156/FSN 148655: Density perturbation in Perturbed dose and Robust eval gives a lower range perturbation	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness
Minor bug fixes	Fixed minor performance issues, such as crashes and low speed, to bring product to its initial specifications.	Substantially Equivalent. The feature does not raise different questions of safety or effectiveness

VII. Non-Clinical and/or Clinical Tests Summary

Software verification and validation testing were conducted, and documentation was provided as recommended by FDA’s Guidance for Industry and FDA Staff, “Guidance for the Content of Premarket Submissions for Software Contained in Medical Devices”. The software for this device was considered as a “Major” level of concern, since a failure or latent flaw in the software could directly result in serious injury or death to the patient.

Cybersecurity and Interoperability requirements were assessed per FDA guidance’s “Cybersecurity in Medical Devices: Quality System Considerations and Content of Premarket Submissions (Sept 2023)”. The cybersecurity analysis showed that the cybersecurity risks are mitigated, and the residual risk is acceptable. The devices are secure for use in their intended environment and methods are in place for ensuring security throughout the total product lifecycle.

Testing of new features followed a multi-level test approach, through unit, integration, and system level testing. Testing to support that new and modified features are comparable to the predicate device was performed through dose engine validation using measurement comparison tests with

Gamma evaluation criteria, plan comparisons using clinical objectives, and through workflow testing. Test results demonstrate conformance to applicable requirements and specifications.

Compliance with device-specific recognized consensus standards, along with general and collateral safety and performance standards for medical devices listed below, ensures that basic safety and essential performance requirements are met.

Standards applied:

Standard No.	Standard Title	Recognition no.
IEC 61217	Radiotherapy equipment - Coordinates movements and scales	12-267
IEC 62304	Medical device software - Software life cycle processes	13-79
IEC 62366-1	Medical devices - Part 1: Application of usability engineering to medical devices	5-129
ISO 14971	Medical devices - Application of risk management to medical devices	5-125
IEC 62083	Medical electrical equipment - Requirements for the safety of radiotherapy treatment planning systems	12-217
IEC 81001-5-1	Health software and health IT systems safety effectiveness and security - Part 5-1: Security - Activities in the product life cycle	13-122

Automated and manual verification activities were created as part of the implementation of product backlog items and performed throughout the project, i.e., continuously during development and during verification and validation preparation and finally during formal verification and validation.

The type of data obtained through software verification is test results from automated and manual test runs. From automated tests, the data type is binary pass or fail result. From manual tests, the type of data is also pass or fail based on manual execution of tests according to specification and comparison of results with the specified expected results.

The purpose of data obtained through verification is to determine successful or failed verification of the requirement linked to the verification activity. The requirement can state e.g. that a certain accuracy must be obtained, a certain input data range is allowed, or that some information must be clearly displayed to the user. The purpose of the verification of the requirements is to verify the consistency, completeness, and correctness of the software and its supporting documentation. The purpose of the verification is also to verify the quality of the source code, correctness of software design, the integration of internal and external components and the correct function of UI components and use cases.

According to the FDA document “General Principles of Software Validation; Final Guidance for Industry and FDA Staff”, software verification “provides objective evidence that the design outputs of a particular phase of the software development life cycle meet all of the specified requirements for that phase”, and software validation is “confirmation by examination and provision of objective

evidence that software specifications conform to user needs and intended uses, and that the particular requirements implemented through software can be consistently fulfilled”.

Below is an overview of the verification and validation activities used to demonstrate substantial equivalence. The specific validation activities for selected significant features are presented in the table below.

- **Unit Testing:** This involves testing individual software requirements to ensure that small sections of the code function as intended in isolation. It helps identify and fix bugs at an early stage.
- **Integration Testing:** This type of testing focuses on verifying that different modules of the software work together as intended. It ensures that the integrated system functions correctly.
- **System Level Testing:** This testing evaluates the entire software system to ensure it meets the specified requirements to validate the overall behaviour of the system.
- **Cybersecurity Testing:** This testing assesses the software's ability to protect against cyber threats and vulnerabilities. It includes penetration testing to ensure the software is secure and resilient against attacks.
- **Usability Testing, also referred to as Validation in a clinical environment:** This testing evaluates the software's user interface and user experience. It involves testing the software with real users to identify usability issues and ensure that the software is as safe as the predicate, easy to use and meets user needs.
- **Regression Testing:** This testing ensures that updates to the software do not introduce new bugs or negatively impact existing functionality. It involves re-running previously conducted tests to verify that the software still performs as expected after modifications.

In the below table, an overview of the verification and validation activities is presented for selected significant features.

Added/updated function in RayStation 2024A SP3, 2024A and 2023B compared to predicate device, RayStation 12A.	Verification and validation data used to demonstrate substantial equivalence	Substantially equivalent?
Dose compensation point computation for Tomo Synchrony	System level verification and validation included dosimetric analysis of motion compensated dose, comparing the effects of using dose compensation point values from RayStation and Accuray's proprietary device Precision. Unit Testing. Purpose: • Verifying that for a given 3D dose volume and a dose threshold, the center point of the volume formed by the voxels with dose above the given threshold is	Yes. The successful validation of this feature demonstrates that the device is as safe and effective as the predicate device.

Added/updated function in RayStation 2024A SP3, 2024A and 2023B compared to predicate device, RayStation 12A.	Verification and validation data used to demonstrate substantial equivalence	Substantially equivalent?
	correctly calculated and that the method used calculates the dose compensation point using the correct dose and threshold. Pass criteria: The calculated values for the center point coordinates are equal to values from the version used in the Accuray validation testing. Calculated dose compensation point values are numerically equal to values obtained from calling the method. This serves as a regression test, demonstrating that the values remain unchanged compared to the version used in the Accuray validation tests. • Verify that the calculated values are exported correctly from RayStation to a DICOM format. 1. Pass criteria: Equality between the calculated and exported coordinate point, 2. Point coordinates only exported for Helical Tomo Synchrony plans, and 3. Point coordinates only exported in the correct DICOM item. • Verify that the calculated values are converted correctly from the DICOM format to the format which is sent to Accuray's system. Pass criteria: The DICOM data coordinate point and the coordinate point in the data structure which is sent to the Accuray system shall be equal. and 2. The point coordinates shall be sent only for the relevant plan types	
Point-dose optimization in brachy plans	Unit level tests, purpose: • Verify that the position from the correct image set is used in point-dose objectives and point dose constraints if there are multiple image sets. Pass criteria: The position used in the point-dose objective/constraint is the position of the point on the planning image and not the position of the point on a second image. • Verify that it is possible to add an optimization objective or constraint with respect to a point and that the added constraint refers to the correct point.	Yes. The successful validation of this feature demonstrates that the device is as safe and effective as the predicate device.

Added/updated function in RayStation 2024A SP3, 2024A and 2023B compared to predicate device, RayStation 12A.	Verification and validation data used to demonstrate substantial equivalence	Substantially equivalent?
	Verify that when adding an optimization objective or constraint to a point, it is possible to select a function type and a dose level for the constraint. Pass criteria: All conditions below are met: When adding an optimization objective/constraint to a point, the constraint is added to the list of optimization functions and that the name of the point in the constraint is the same as the name of the selected point. When adding an optimization objective/constraint to a point, the description of the added optimization constraint has the same function type and dose level as was selected. - Verify that it is possible to save and load an optimization function template containing optimization objectives and constraints to points. Pass criteria: After saving and loading a template containing optimization objectives and constraints to points, the loaded optimization functions are the same as the optimization functions saved in the template. - Verify that the results from point optimization is as expected, by optimizing with a single objective for the dose at a point without any other objectives or constraints. Verify that the results from point optimization is as expected, by optimizing with multiple objectives for the dose at different points without any other objectives or constraints. Pass criteria: Both conditions below are met: - After optimization, the dose in the point should be equal to the dose specified in the objective. - After optimization, the dose in all points with an objective should be equal to the dose specified in each objective.	

Added/updated function in RayStation 2024A SP3, 2024A and 2023B compared to predicate device, RayStation 12A.	Verification and validation data used to demonstrate substantial equivalence	Substantially equivalent?
Electron Monte Carlo dose engine improvements	Dose engine validation, comparing calculated doses with measured doses obtained from clinics, doses computed in independent, well-established TPS, doses computed with earlier versions of RayStation, and doses computed in BEAMnrc/egs++ with Gamma evaluation criteria.	Yes. The successful validation of this feature demonstrates that the device is as safe and effective as the predicate device.
Segment weight optimization using photon Monte Carlo added	System-level performance of the optimization method proposed in the subject device was compared with the predicate through resulting plan dose and dose statistics for auto breast planning, SMLC, and VMAT plans.	Yes. The successful validation of this feature demonstrates that the device is as safe and effective as the predicate device.
Photon Monte Carlo dose engine: Improved positron handling	Dose engine validation with Gamma evaluation criteria. After recommissioning of the beam model, the dose differences between photon Monte Carlo computed doses in RayStation 12A and RayStation 2024A are negligible. The dose engine validation shows the same level of accuracy as before after remodeling.	Yes. The successful validation of this feature demonstrates that the device is as safe and effective as the predicate device.
Evaluation on converted CBCT images for protons	Test cases consist of CBCTs from the MedPhoton imaging ring on a Mevion S250i system, as well as the on-board CBCT systems on a Varian ProBeam and an IBA P1. Test cases cover validation of 3D dose computed on both the Corrected CBCT and Virtual CT. For each case, a ground truth CT image has been prepared to serve as ground truth. Test criteria: Gamma 2%/2mm pass rate above 90% for proton MC/PB dose computation Gamma 3%/3mm pass rate above 95% for proton MC/PB dose computation	Yes. The successful validation of this feature demonstrates that the device is as safe and effective as the predicate device.

The data obtained from the verification show that system tests, use error tests, unit and subsystem tests are acceptable, and the validations have been completed successfully. The reviews of design, code and labeling met the acceptance criteria.

The validation of the applicable new/updated features, performed in a clinical environment, were successful. From the user responses to the questions if workflows/functionality tests were successful, if documentation was understandable and if RayStation was considered safe to use, the determination was made that validation of the applicable features passed the criteria for a successful

result and that the RayStation/RayPlan 2024A SP3, 2024A and 2023B safety and effectiveness as compared to the predicate has been validated for clinical use.

has been validated for clinical use.

From the successful verification and validation activities, the conclusion can be drawn that RayStation/RayPlan 2024A SP3, 2024A and 2023B have met specifications and are as safe, as effective and perform as well as the legally marketed predicate devices.

VIII. Animal Testing

No animal testing was required to demonstrate substantial equivalence.

IX. Clinical Testing

No Clinical trials were required to demonstrate substantial equivalence.

X. Conclusion

The differences between the predicate and subject device do not raise any new or different questions of safety or effectiveness.

In the Technological Comparison section above, added/updated functions in RayStation/RayPlan 2024A SP3, 2024A and 2023B are compared with the predicate device, RayStation 12A in a General features comparison table and in an Added/updated functions table. For both tables, all compared items are evaluated as substantially equivalent and do not raise different questions of safety of effectiveness.

The non-clinical data support the safety of the device as compared to the predicate and the software verification and validation demonstrate that the RayStation/RayPlan 2024A SP3, 2024A and 2023B devices should perform as intended in the specified use conditions, and the performance testing demonstrates that the RayStation/RayPlan 2024A SP3, 2024A and 2023B devices perform as well as the predicate device that is currently marketed for the same intended use.

Therefore, all of the differences are substantially equivalent compared with the predicate device.

RayStation/RayPlan 2024A SP3, 2024A and 2023B are substantially equivalent to RayStation 12A cleared under K222312.